

Heterogeneous Learning Systems: A Relevance Cascade for Collective Capability and Competence Evolution

Francisco Javier González Serrano
Universidad Carlos III de Madrid

Working paper v0.2 — October 2026

Abstract

Heterogeneous Learning Systems (HLS) concerns collectives whose limited, distributed competences must be organized now and may evolve through what the collective does. Its central scientific claim is neither that heterogeneity, learning, nor integrated control is universally valuable. It is that a proposed organization–development effect must pass a sequence of distinct relevance filters before it can alter a present choice. We call this sequence the HLS Relevance Cascade:

$$a_t \longrightarrow S_{t+1} \longrightarrow C_{t+1} \longrightarrow V_{t+1} \longrightarrow \text{optimal present choice.}$$

The cascade separates state divergence (F1), capability relevance (F2), value relevance (F3), and decision relevance (F4), with $F4 \Rightarrow F3 \Rightarrow F2 \Rightarrow F1$ and generally false converses. It is a structural HLS screen, not a new dynamic-programming theorem. We combine a static collective-capability operator with an endogenous competence transition, reuse the former on the future state, and state the minimal consistency condition connecting capability to value. Established assignment, learning, and continuation-value mathematics are deliberately imported. The scientific contribution lies in the HLS architecture, the distinctions it requires, their null boundaries, and their assembly into one testable collective problem. Static G1 controls establish the Beam-1 distinctions; dynamic G2 provides a staged witness for F1–F4. The result is a compact foundation for deciding when a richer HLS model is warranted and when the problem reduces to a simpler one.

1 The scientific claim

Collective problem solving is often organized around heterogeneous units: human workers, service teams, autonomous agents, or portfolios of learned models can differ in competence, cost, capacity, and future development. HLS studies the system-level problem created when such a collective must both use its current competences and determine which present actions affect its future competences. The motivating questions are simple: *who should do what?* and *who should learn what, how, and from whom?*

The contribution is not a claim that a heterogeneous, learning, or jointly managed system dominates a simpler alternative. Nor is it a replacement for assignment theory, learning theory, or dynamic programming. The claim is more specific: these established components address different objects, and an HLS analysis must preserve their order if it is to tell whether a physical competence change can matter for a collective decision. The relevant object is therefore the *architecture of dependence* from realized action to future state, collective capability, future value, and present choice.

This framing has two consequences. First, an equality or a null is informative: it locates the first point at which a proposed HLS effect disappears. Second, new machinery is unwarranted when the cascade stops early. A fixed-competence problem can be treated as static organization; a state change that leaves collective capability unchanged need not be promoted to a dynamic organization problem; and a future-value difference that leaves the present optimum unchanged does not establish decision relevance. This is a scientific criterion for reduction, not a generic preference for more elaborate models.

2 Beam 1: collective capability under fixed competence

The static beam asks what a collective can achieve from its current competence structure. Let S denote the competence state, P the problem or demand conditions, B the operational constraints, and Π an admissible class of organizational policies. For fixed (S, P, B, Π) , the feasible collective outcome set is

$$\mathcal{F}_{\Pi}(S, P, B) = \{z^{\pi}(S, P) : \pi \in \Pi_B\}, \quad (1)$$

and the non-dominated collective-capability frontier is

$$\mathcal{C}_{\Pi}(S, P, B) = ND[\mathcal{F}_{\Pi}(S, P, B)]. \quad (2)$$

This is the static T0 object. It separates competence from feasible collective outcomes, collective capability, realized performance, and value. Organization is deliberately broader than routing: it is the admissible way of accessing and using distributed competence.

This formulation imports familiar static machinery rather than recasting it as new HLS mathematics. The organization of distributed knowledge [4], assignment and specialization, and resource-constrained allocation already address important parts of Beam 1. HLS supplies the common semantic location in which their outcomes can be compared before and after competence evolution.

For nested organizational classes, a richer class weakly expands the attainable set. That inclusion alone is not the scientific result: it may leave the efficient frontier unchanged (organizational sufficiency), and an expanded frontier may still have no realized value under the specified P and B . Thus the Beam-1 distinction is

$$\text{heterogeneity} \not\Rightarrow \text{attainable expansion} \not\Rightarrow \text{efficient capability expansion} \not\Rightarrow \text{organizational value}. \quad (3)$$

The arrows in (3) are a sequence of questions, not a claim that the intermediate objects are interchangeable.

G1: the static reference. G1 is a fixed-competence, two-agent/two-task ground truth for these distinctions. It includes homogeneous and dominance nulls, crossed specialization, resource-effectiveness trade-offs, and a budget control in which the same competence structure has zero, then positive, then zero organizational value. It therefore establishes only the scoped Beam-1 lesson: organizational value is a property of (S, P, B, Π) , not of S or heterogeneity alone. G1 is not evidence about competence evolution or a claim of novel static operations-research theory.

3 The HLS Relevance Cascade

3.1 From realized action to future capability

The minimum dynamic boundary is endogenous competence change through realized organizational use:

$$S_{t+1} = F(S_t, a_t). \quad (4)$$

The distinction between the policy and the realized action matters: the transition is caused by what the collective does, not merely by the nominal policy description. If the future state is independent of present action, the system is a succession of static Beam-1 problems.

Dynamic HLS does not define a second kind of collective capability. It reuses (2) under future operating conditions:

$$\mathcal{C}_{t+1}(a_t) = \mathcal{C}_{\Pi_{t+1}^*}(F(S_t, a_t), P_{t+1}, B_{t+1}). \quad (5)$$

For alternatives a, a' , this permits a first distinction between a changed state and changed efficient collective capability.

Let $V_{t+1}(S; \theta_{t+1})$ be the best future collective performance under common future operating conditions $\theta_{t+1} = (P_{t+1}, B_{t+1}, \Pi_{t+1})$ and the performance criterion of the instantiated problem. No universal scalar HLS utility is assumed. The minimal T4→T5 consistency condition is:

$$\mathcal{C}_{t+1}(a) = \mathcal{C}_{t+1}(a') \implies V_{t+1}(F(S_t, a); \theta_{t+1}) = V_{t+1}(F(S_t, a'); \theta_{t+1}), \quad (6)$$

provided future conditions are common and the criterion is compatible with T0, that is, evaluated from the same collective-capability object. Equality of capability is therefore a sufficient null for comparative future value under those conditions; unequal capability need not yield unequal value.

3.2 Four successive filters

For two present alternatives under common future conditions, the HLS Relevance Cascade is

1. **F1 — state divergence:** $F(S_t, a) \neq F(S_t, a')$.
2. **F2 — capability relevance:** $\mathcal{C}_{t+1}(a) \neq \mathcal{C}_{t+1}(a')$.
3. **F3 — value relevance:** the corresponding future values differ under the common future conditions.
4. **F4 — decision relevance:** for a two-alternative present choice with a strict static preference, the future-value term changes the optimal present choice.

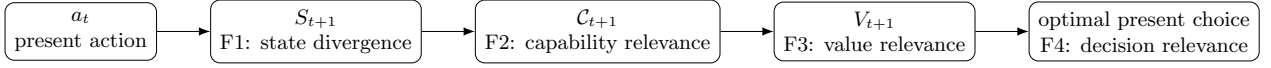


Figure 1: The HLS Relevance Cascade. Each filter asks whether the preceding difference survives in the next HLS object.

Control	Cascade status	Scoped lesson
D0	F1 blocked	No evolution: alternative actions induce the same future state, capability, and value.
D1	F1 passes; F2 blocked	Learning changes state and attainable outcomes, but both efficient frontiers are (0.8, 0.8).
D2, $p = 0$	F2 passes; F3 blocked	Future efficient capabilities differ, but $\Delta^{\text{dev}}(0) = 0$.
D2, $0 < p < 0.5$	F3 passes; F4 not reached	$\Delta^{\text{dev}}(p) > 0$, yet it does not overcome the present operational disadvantage.
D2, $p > 0.5$	F4 reached	Future development value reverses the strictly static preference.

Table 1: G2 as a staged witness for the HLS Relevance Cascade.

The structural implication is

$$\boxed{F4 \Rightarrow F3 \Rightarrow F2 \Rightarrow F1.} \quad (7)$$

If F1 fails, the future states and hence capability and value coincide. If F2 fails, (6) blocks F3. If F3 fails, no differential continuation value can change a fixed present ranking. The converses are false in general: a state difference can be collectively redundant; different capabilities can be equally valuable; and a positive value difference can be too small to reverse a strict static preference.

For the minimal two-period comparison, let $\Delta R_t(a, a')$ be the present operational difference and $\Delta_{t+1}^{\text{dev}}(a, a')$ the future-value difference. Then

$$\Delta J_t = \Delta R_t + \Delta_{t+1}^{\text{dev}}. \quad (8)$$

This is ordinary continuation-value reasoning. The cascade is not a theorem of dynamic programming and does not claim new optimization mathematics; it is the HLS structural screen that says which competence–organization distinctions must be retained before (8) is scientifically relevant.

4 G2 as a staged witness

G2 is a deliberately minimal two-agent, two-task, two-period constructive and null reference. It uses deterministic learning by doing only to instantiate F ; it is neither a universal dynamic HLS model nor an empirical campaign. Its value is that each control stops, or completes, the same cascade in Fig. 1.

In D2, the present difference and development value are

$$\Delta R = -0.1, \quad \Delta^{\text{dev}}(p) = 0.2p, \quad \Delta J(p) = -0.1 + 0.2p. \quad (9)$$

Consequently, $p = 0$ is the T5 null, $0 < p < 0.5$ exhibits value relevance without decision relevance, $p = 0.5$ is intertemporal indifference, and $p > 0.5$ reaches F4. This is precisely why passing F1–F3 is not sufficient for F4. G2 supplies a witness for the logical boundaries; it does not establish the prevalence, robustness, or irreducibility of any dynamic organizational arrangement.

5 What the HLS architecture adds

The mathematics used at each stage is intentionally familiar. Static collective capability can draw on assignment, resource-allocation, and organizational theories of distributed knowledge [4]. Transitions can instantiate established accounts of learning by doing, task-specific human capital, or organizational learning [1, 2, 5]. Once a particular state, transition, and criterion have been specified, dynamic programming supplies the appropriate continuation-value calculation. Competence-oriented dynamic portfolio work and dynamic staffing already study particular operation–development interactions [3, 6].

HLS does not relabel these components as a new mathematical theory. Its scientific contribution is their disciplined assembly around a collective problem: (i) one static capability semantics before and after competence evolution; (ii) realized action as the causal interface to future competence; (iii) an explicit separation of state, capability, value, and decision relevance; and (iv) null boundaries that identify when the composition reduces

to a simpler established problem. These are not merely a catalogue of framework elements. Together they impose falsifiable structural requirements on any claim that organization and competence development are consequential.

The architecture also disciplines interpretation. F4 does not imply that centralized or irreducibly joint management is required: a sufficient continuation signal or a modular decomposition may reproduce the same optimum. Conversely, a failure before F4 is not an unsuccessful HLS experiment. It is a scoped result showing which proposed dependency is absent under the stated conditions. HLS therefore treats reduction, sufficiency, and null results as scientifically informative outcomes.

6 Scope and conclusion

The foundation makes no claim of universal learning value, universal heterogeneity value, universal superiority of dynamic organization, novel dynamic-programming theory, or necessity of irreducibly joint management. It does not introduce a learning mechanism, an algorithm, a new dynamic block, or a universal scalar objective. Such additions require a prior HLS question and must be assessed against the same relevance screen.

The central result is a compact scientific architecture for a collective with heterogeneous and evolvable competences. Beam 1 establishes that competence, collective capability, and realized organizational value are distinct. The dynamic beam adds the minimum endogenous transition and reuses the same capability operator. The HLS Relevance Cascade then makes the required successive tests explicit: a present action must create a state difference, that difference must affect collective capability, capability must affect future value, and future value must affect the optimal present choice. G1 and G2 provide minimal null and positive controls for this argument. The resulting foundation states both when an HLS effect is substantively present and when no richer HLS machinery is warranted.

References

- [1] Linda Argote and Ella Miron-Spektor. Organizational learning: From experience to knowledge. *Organization Science*, 22(5):1123–1137, 2011. doi: 10.1287/orsc.1100.0621.
- [2] Kenneth J. Arrow. The economic implications of learning by doing. *The Review of Economic Studies*, 29(3):155–173, 1962.
- [3] Tessa Borgonjon and Broos Maenhout. The impact of dynamic learning and training on the personnel staffing decision. *Computers & Industrial Engineering*, 187:109784, 2024. doi: 10.1016/j.cie.2023.109784.
- [4] Luis Garicano. Hierarchies and the organization of knowledge in production. *Journal of Political Economy*, 108(5):874–904, 2000.
- [5] Robert Gibbons and Michael Waldman. Task-specific human capital. *American Economic Review*, 94(2): 203–207, 2004.
- [6] Walter J. Gutjahr. Optimal dynamic portfolio selection for projects under a competence development model. *OR Spectrum*, 33:173–206, 2011. doi: 10.1007/s00291-009-0180-9.